

prices have now reduced by almost ninety per cent due to the aggressive launch of self-driving cars by Google's Waymo and others. Also, 3D cameras that used to be very expensive are now within the reach of every company.

One of the leading international business research firms, Freedonia Group, has speculated that the worldwide escalation for robots will be around eleven per cent each year in the coming three years. The huge improvement in technology during the past ten years has helped enhance the growth of robotics.

Currently, robots are mainly used in material handling, assembling, and welding applications in industries. According to a survey by McKinsey & Company published last year, the industrial robotics market experienced record unit sales of 381,000 in 2017, and currently has an installed base of about 2.1 million units worldwide. The market for industrial robots has been growing at a record rate of around nineteen per cent per annum since 2012, and is expected to continue like this until 2021.

Highlighting the development, growth, and usefulness of robotics, Balaji Viswanathan, founder and CEO of Invento Robotics, says, "In large auto warehouses, robots now run the major portion of the operation that doesn't need much light, thus reducing the cost of operation. Right now, our virtual assistant enabled chatbots are also a kind of robot that solves our various queries online."

Usefulness and growth of robotics in global business

Previously, the interaction between a man and machine was very limited and monotonous. But now, with the introduction of new robots, interactions have become very interesting. Nowadays, these robots interact and assist in various physical activities, and also have the ability to save our lives.

Various organisations around the world have now realised the importance of robots and taken important steps to start using these in their various applications. For example, MIT has unveiled a soft robotic fish known as SoFi that can help save the vast ocean. NASA is all set to send a huge

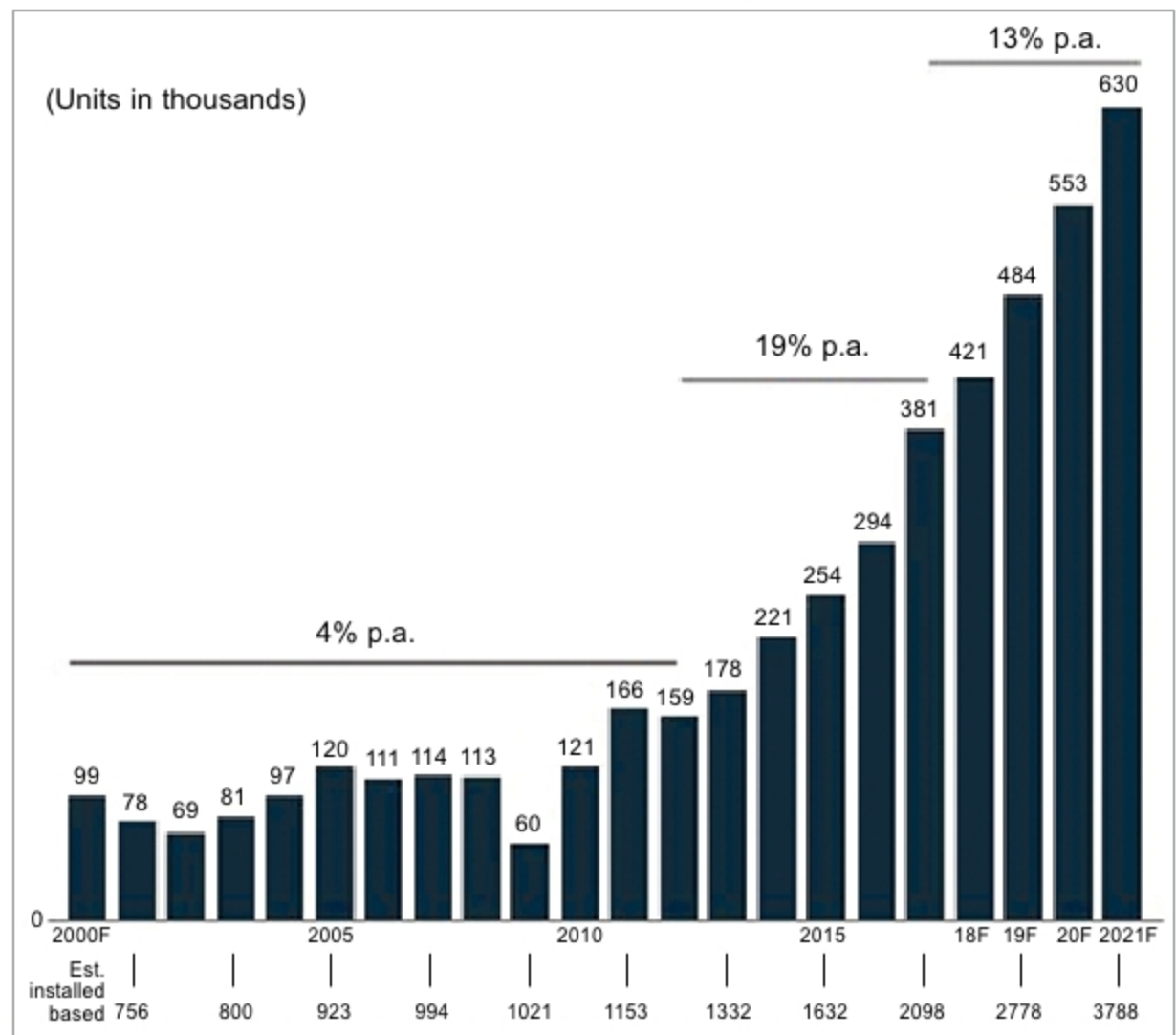


Fig. 1: Annual shipments of industrial robots worldwide

bunch of robots to Mars.

Not only this, robotics is also playing a huge role in the giant automobile industry. For instance, in a BMW facility in Chennai, robotic arms pick up material, thus helping in the smooth operation of the assembly line. While such mechanical jobs done by a human involve a lot of effort, robots make these jobs extremely easy and trouble-free. Most importantly, robots make the assembly and manufacture more consistent and efficient.

According to a study by Research Dive, the global collaborative robot (cobot) market will surpass 8,840.5 million dollars by 2026, at a CAGR of 41.2 per cent. Various industries are increasingly adopting cobots due to such advantages as increased productivity, effective employee utilisation, and improved workplace safety.

Phanindra Karody, head, Bengaluru plant, Continental Automotive India, says, "Cobots are designed to share the workspace with humans and take over duties that can result in being burdening, hazardous, or unergonomic such as assembling and carrying heavy components. Employing cobots to take over such mundane work results in minimal or no error

while keeping the downtime in check. On the other hand, workers can focus on tasks requiring close attention and maintaining precision at production, quality, and testing. Thus, promoting productivity, quality, efficiency, and sustainability (in terms of cost-saving, resource-saving, and safety) on the shop floor."

The robotics firm Addverb exclusively stated to EFY that robots and cobots (collaborative robots) have emerged as a preferred choice by businesses to acclimatise their manufacturing with these changing times. With the onset of Covid-19 pandemic, robotics and increasing automation have become a strategic priority for global businesses.

If implemented correctly, robotics can not only contribute to the smooth running of the businesses but also enable scalability. The technology can ensure consistency so that the same level of quality is achieved from company-wide operations. Flexibility, agility, and productivity are three key drivers that are primarily accelerating the adoption of cobots. With the change in the supply chain paradigm and the quest to provide customers with the best product in the shortest